Program Registration & Attendance Reconciliation Hub

Focused Five-Page Proposal | Fall 2026 Software Engineering Capstone

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CAPSTONE FORMAT Approximately four Master of Software Engineering students	**PROJECT PERIOD** September 24-December 4, 2026 (10 weeks)

Project summary

UCI Graduate Division proposes a secure, file-based web application that helps staff combine event registration records with actual attendance evidence. The prototype will import a limited set of representative spreadsheets, reconcile participant identities, route uncertain matches to staff for review, support both one-time events and multi-session certificate programs, and produce consistent reports and exports. The project addresses the data-cleanup and identity-resolution work that must occur before a dependable dashboard or broader institutional integration can be built.

The problem

Registration is commonly collected in 12twenty or CampusGroups/ZotSpot, while attendance may come from Zoom reports or in-person sign-in sheets. Staff manually compare these sources to determine who registered, attended, did not attend, or appeared without registering. Zoom display names and personal email addresses often differ from institutional records, creating duplicates and uncertain matches. Certificate programs add approved cohorts, multiple sessions, occasional additional requirements, and a separate decision about who was ultimately awarded a certificate.

Project objective

1. Create a trustworthy, reviewable attendance record from a small set of known exports.
2. Reduce repetitive spreadsheet comparison while keeping staff responsible for ambiguous identity decisions.
3. Provide useful event and certificate-program reports without direct student-information-system access.
4. Deliver a demonstrable foundation that can be extended in a later phase.

02 Required MVP and Staff Workflow

The required product is a working prototype using defined file formats, not a production enterprise system.

Supported input files

SOURCE	REQUIRED MVP USE
12twenty	Import a known CSV/XLSX registration export containing participant, program, registration, and attendance-related fields.
CampusGroups/ZotSpot	Import a known registration export containing contact information, IDs where available, registration status, and event identifiers.
Zoom	Import a participant report as attendance evidence; retain display name, email, join/leave, and duration fields when available.
In-person sign-in	Import one standardized spreadsheet template for actual attendance.
Certificate cohort	Import a standardized list of selected participants; arbitrary interest-form interpretation is not required.

Core workflow

5. Upload registration or cohort data for an event or program.
6. Upload Zoom or in-person attendance evidence for the corresponding session.
7. Normalize source columns into common participant, event, registration, and attendance records.
8. Match exact identifiers first: NetID, student/employee ID, and normalized email.
9. Present possible name or email matches with a confidence indicator; never automatically merge an uncertain match.
10. Let staff confirm the same person or keep records separate, then finalize attended, absent, walk-in, or unresolved status.
11. Generate reports and export reconciled data for authorized users.

Identity review

The matching feature should be intentionally modest. Exact identifiers produce direct matches. When those are missing, the system may rank a small set of candidates using normalized names and emails. Staff make the final decision. Full record merging, splitting, and cross-event alias management are stretch features rather than required deliverables.

03 Program Models and Reporting

The same core data structure must support simple events and longitudinal certificate participation.

Two program models

ONE-TIME EVENT	CERTIFICATE OR MULTI-SESSION PROGRAM
A registration roster is compared with one attendance source.	A selected cohort is connected to several sessions and may include drop-in attendees.
Final statuses: attended, absent, walk-in, or unresolved.	Track cohort status, session attendance, and optional staff-entered requirement notes.
Report registrations, attendance, no-shows, walk-ins, and exceptions.	Report progress and completion; certificate-awarded status remains a separate staff-confirmed field.

Required functional capabilities

12. Create events, programs, sessions, and participant records.
13. Preserve registration and attendance as separate evidence rather than overwriting one with the other.
14. Identify missing identifiers, duplicate-looking records, unmatched attendance, and unresolved decisions.
15. Distinguish selected certificate participants from drop-in session attendees.
16. Allow staff to record whether additional requirements were met and separately confirm certificate award.
17. Retain source-file and import information sufficient to explain where a record came from.
18. Permit unavailable identifiers, demographic, and academic fields to remain blank without inferring a characteristic.

Required reports and exports

OUTPUT	PRIMARY CONTENT	USE
Event reconciliation	Registered, attended, absent, walk-in, unresolved	Finalize an event and correct exceptions
Certificate progress	Cohort status, attendance by session, award status	Monitor progress and completion
Program reach	Unique participants and repeat participation	Summarize service reach
Data quality	Missing IDs, possible duplicates, unmatched records	Prioritize staff cleanup

04 Scope Boundaries, Security, and Success Criteria

Clear exclusions protect the ten-week schedule and make completion measurable.

Scope boundaries

REQUIRED MVP	STRETCH GOALS	OUT OF SCOPE
Known file imports and a common data model	Institutional data-request workbook generator	Direct student-information-system integration
Exact matching plus staff-reviewed suggestions	Re-import of returned enrichment files	Live vendor APIs or arbitrary spreadsheet parsing
One-time and basic certificate tracking	Postdoc roster enrichment	UCI SSO, production hosting, or production deployment
Core reports, exports, and exception views	Cross-event aliases and richer audit history	Automatic fuzzy merges or predictive analytics
Basic application login and user roles	Additional dashboards and visualizations	Configurable certificate-rules engine

Security and privacy requirements

19. Only authenticated users may view or export participant information.
20. At minimum, support administrator/program-staff and read-only/reporting roles.
21. Apply least-privilege access and record basic import and identity-review actions.
22. Use synthetic or de-identified files during the capstone; do not use live protected student data.
23. Treat production use, institutional authentication, hosting, retention, and data access as later UCI privacy and security reviews.

Measurable acceptance criteria

24. Staff can import each required source format and see validation errors for unusable files.
25. The system performs exact matching and routes uncertain candidate matches to staff review.
26. Registration and attendance evidence remain separate and produce the four required final event statuses.
27. A demonstration covers both a one-time event and a multi-session certificate cohort.
28. Authorized users can generate the required reports and a reconciled export.
29. Role restrictions prevent a read-only user from importing or changing records.

05 Deliverables, Timeline, and Partnership

The project is structured as an end-to-end proof of concept with a documented path to future expansion.

Expected deliverables

30. Working web prototype with documented setup instructions.
31. Database schema and data dictionary for participants, events, registrations, sessions, attendance, and awards.
32. Import templates, field mappings, validation behavior, and synthetic sample files.
33. Identity-matching and staff-review workflow with test cases.
34. Event, certificate, reach, and data-quality reports plus downloadable reconciled data.
35. User guide, automated tests where practical, known-limitations list, and technical handoff documentation.

Proposed ten-week plan

WEEKS	FOCUS	CHECKPOINT
1-2	Confirm workflows, file mappings, data model, and interface plan	Approved requirements and prototype design
3-4	Build imports, validation, normalization, and core records	Representative files load successfully
5-6	Implement exact matching, candidate suggestions, and staff review	End-to-end reconciliation demonstration
7-8	Add certificate tracking, reports, exports, login, and roles	Feature-complete MVP
9-10	Test, refine, document, prepare handoff, and present	Final demonstration and deliverables

Partner support and resources

UCI Graduate Division will provide representative de-identified or synthetic versions of registration, attendance, cohort, and reporting files; explain current workflows; answer questions; and provide regular, timely feedback. The project partner will participate in required orientation, attend periodic check-ins, review demonstrations, help resolve workflow questions, and participate in the final showcase. The MSWE program can provide technical supervision.

Feasibility and future path

This focused scope is reasonable for approximately four students over ten weeks because it limits integrations, formats, identity actions, and reports while preserving a meaningful full-stack challenge. If the MVP finishes early, the preferred first stretch goal is a data-request workbook listing participant identifiers and blank institutional fields. Direct integrations, richer dashboards, and production deployment belong to later phases.